Tutorial Notes for AIST3010

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1 Introduction

In this course, we focus on the following optimization problems

$$\begin{aligned} &\min_{\mathbf{x}} f(\mathbf{x}) \\ &\text{Subject to } \ \mathbf{x} \in \mathcal{X}, \end{aligned} \tag{1}$$

where we call f the object function and $\mathcal{X} \subseteq \mathbb{R}^d$ is the feasible set.

2 Basic Matrix Analysis

This section provide some techniques in matrix analysis. To learn more about matrix, please refer to matrix cookbook Petersen et al. [2008 http://www2.imm.dtu.dk/pubdb/edoc/imm3274.pdf].

2.1 Norm

Definition 2.1 (l_p**-norm**). *for* $p \geq 1$

$$\|\mathbf{x}\|_p = (|x_1|^p + |x_2|^p + \cdots + |x_d|^p)^{\frac{1}{p}} \tag{2}$$

where

$$\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_d \end{bmatrix}. \tag{3}$$

Proposition 2.2. $\|\mathbf{x}\|_\infty \overset{def}{=} \lim_{p\to\infty} \|\mathbf{x}\|_p$, *which is equal to*

$$\|\mathbf{x}\|_\infty = \max_{1\leq i\leq d} |\mathbf{x}_i| \tag{4}$$

Proof. Hint: W.O.L.G, we assume $0 < |x_1| = \max_{1\leq i\leq d} |x_i|$

$$\begin{aligned} \|\mathbf{x}\|_\infty = \lim_{p\to\infty} \|\mathbf{x}\|_p &= \lim_{p\to\infty} (|x_1|^p + |x_2|^p + \cdots + |x_d|^p)^{\frac{1}{p}} && (5) \\ &= \lim_{p\to\infty} \left(|x_1|^p \left(1 + \frac{|x_2|^p}{|\mathbf{x}_1|^p} + \cdots + \frac{|x_d|^p}{|x_1|^p} \right) \right)^{\frac{1}{p}} && (6) \\ &= |x_1| \lim_{p\to\infty} \left(1 + \frac{|x_2|^p}{|\mathbf{x}_1|^p} + \cdots + \frac{|x_d|^p}{|x_1|^p} \right)^{\frac{1}{p}} && (7) \end{aligned}$$

Then use squeeze theorem. □

Proposition 2.3.

$$1 \le p \le q \le \infty \Rightarrow \|\mathbf{x}\|_p \ge \|\mathbf{x}\|_q \tag{8}$$

Proof. Hint: Construct the function: $f(r) = (|x_1|^r + \cdots + |x_d|^r)^{1/r}$, and prove that $f'(r) \le 0$ for $r \in [1, \infty]$.[1] □

We then focus on l_1, l_2 and l_∞ norm

Proposition 2.4. *We have*

$$\|\mathbf{x}\|_2 \le \|\mathbf{x}\|_1 \le \sqrt{d}\|\mathbf{x}\|_2 \tag{9}$$

Proof. $\|\mathbf{x}\|_2 \le \|\mathbf{x}\|_1$ can be directly obtained by Proposition 2.3. The right half inequality can be established by using Cauchy–Schwarz inequality that is

$$\|\mathbf{x}\|_1 = 1 \cdot |x_1| + \cdots + 1 \cdot |x_d| \le \sqrt{\underbrace{1 + \cdots + 1}_{d}}\sqrt{\sum_{i=1}^{d} |x_i|^2} = \sqrt{d}\|\mathbf{x}\|_2 \tag{10}$$

□

Questions

1. For every vector $\mathbf{x}, \mathbf{y} \in \mathbb{R}^d$, we have $\mathbf{x}^\top \mathbf{y} \le \|\mathbf{x}\|_1 \|\mathbf{y}\|_1$? (True)
2. For every vector $\mathbf{x}, \mathbf{y} \in \mathbb{R}^d$, we have $\mathbf{x}^\top \mathbf{y} \le \|\mathbf{x}\|_\infty \|\mathbf{y}\|_\infty$? (False)

2.2 PSD and PD

We only focus on the matrix in the real space.

Definition 2.5 (Symmetric matrix). $\mathbf{H} = \mathbf{H}^\top$, *we call* $\mathbf{H}$ *is symmetric.*

Proposition 2.6. *For every real symmetric matrix* $\mathbf{H}$ *and* $\operatorname{rank}(\mathbf{H}) = r$, *there exists an orthogonal matrix* $\Gamma \in \mathbb{R}^{r \times d}$ *where* $\Gamma^\top = [\mathbf{y}_1, \mathbf{y}_2, \cdots, \mathbf{y}_r]$ *and a diagonal matrix* $\Sigma = \operatorname{diag}(\lambda_1; \cdots; \lambda_r) \in \mathbb{R}^{r \times r}$ *such that*

$$\mathbf{H} = \Gamma^\top \Sigma \Gamma \tag{11}$$

Remark 2.7. *We can always find an orthogonal matrix* $\bar{\Gamma} \in \mathbb{R}^{d \times d}$ *s.t.* $\bar{\Gamma}^\top = [\Gamma^\top, \mathbf{y}_{r+1}, \cdots, \mathbf{y}_d]$ *and*

$$\mathbf{H} = \bar{\Gamma}^\top \bar{\Sigma} \bar{\Gamma}, \tag{12}$$

where $\bar{\Sigma} = \operatorname{diag}(\lambda_1; \cdots; \lambda_r; 0; \cdots; 0) \in \mathbb{R}^{d \times d}$. *We always use this type of decomposition when doing matrix analysis.*

Definition 2.8 (Postive-(Semi)-Definite Matrix). $\mathbf{H} \in \mathbb{R}^{d \times d}$ *is symmetric, we call* $\mathbf{H}$ *positive-semidefinite (*$\mathbf{H} \succeq \mathbf{0}$*) if:*

$$\mathbf{x}^\top \mathbf{H} \mathbf{x} \ge 0, \quad \textit{for all} \;\; \mathbf{x} \ne \mathbf{0}, \;\; \mathbf{x} \in \mathbb{R}^d. \tag{13}$$

We call $\mathbf{H}$ *positive-definite (*$\mathbf{H} \succ \mathbf{0}$*) if*

$$\mathbf{x}^\top \mathbf{H} \mathbf{x} > 0, \quad \textit{for all} \;\; \mathbf{x} \ne \mathbf{0}, \;\; \mathbf{x} \in \mathbb{R}^d. \tag{14}$$

[1] For the details, you can check: `https://math.stackexchange.com/questions/1946511`.

Proposition 2.9. *If* $\mathbf{H}$ *is a symmetric matrix, then the following statements*

- *(a).* $\mathbf{H} \succeq \mathbf{0}$
- *(b).* $\mathbf{H}$*'s eigenvalues are all non-negative*
- *(c). There exits a matrix* $\mathbf{A}$ *s.t.* $\mathbf{H} = \mathbf{A}^\top \mathbf{A}$

Proof. We can prove this by the loop

$$\text{(c)} \Rightarrow \text{(a)} \Rightarrow \text{(b)} \Rightarrow \text{(c)} \tag{15}$$

□

Proposition 2.10. *If* $\mathbf{H}$ *is a symmetric matrix, then the following statements*

- *(a).* $\mathbf{H} \succ \mathbf{0}$
- *(b).* $\mathbf{H}$*'s eigenvalues greater than 0*
- *(c). There exits a matrix* $\mathbf{A} \succ \mathbf{0}$ *s.t.* $\mathbf{H} = \mathbf{A}^\top \mathbf{A}$

Proof. We can prove this by the loop

$$\text{(c)} \Rightarrow \text{(a)} \Rightarrow \text{(b)} \Rightarrow \text{(c)} \tag{16}$$

□

Questions

1. Every PSD matrix $\mathbf{A} \in \mathbb{R}^{d \times d}$ is full-rank? (False)
2. Every PD matrix $\mathbf{A} \in \mathbb{R}^{d \times d}$ is full-rank? (True)
3. If $\mathbf{A} \succ \mathbf{0}$, then $\mathbf{A}^{-1} \succ \mathbf{0}$ (True)
4. If $\mathbf{A} \succ \mathbf{B} \succ \mathbf{0}$, then $\mathbf{A}^2 \succ \mathbf{B}^2$ (False)*
5. If $\mathbf{A} = \begin{bmatrix} \mathbf{A}_{11} & \mathbf{A}_{12} \\ \mathbf{A}_{21} & \mathbf{A}_{22} \end{bmatrix} \succ \mathbf{0}$, then $\mathbf{A}_{11} \succ \mathbf{0}$ and $\mathbf{A}_{22} \succ \mathbf{0}$ (True)

2.3 Matrix Norm*

In this section, we introduced the matrix norm.

Definition 2.11 (Matrix Norm (Induced by Vector Norm)).

$$\|\mathbf{A}\|_{p,q} \stackrel{def}{=} \sup_{\mathbf{x} \neq \mathbf{0}} \frac{\|\mathbf{A}\mathbf{x}\|_q}{\|\mathbf{x}\|_p} = \sup_{\mathbf{x} \neq 0,\ \|\mathbf{x}\|_p = 1} \|\mathbf{A}\mathbf{x}\|_q, \tag{17}$$

where $\|\cdot\|_p$, $\|\cdot\|_q$ *are the defined vector norms in Definition 2.1.*

When $p = q$ in (17), we have the p-norm of matrix:

Definition 2.12 (**Matrix p-Norm**).

$$\|\mathbf{A}\|_p \overset{def}{=} \|\mathbf{A}\|_{p,p} = \sup_{\mathbf{x}\neq\mathbf{0}} \frac{\|\mathbf{A}\mathbf{x}\|_p}{\|\mathbf{x}\|_p} \tag{18}$$

We focus on matrix 2*-norm and provide the following proposition*

Proposition 2.13. *Consider the* 2*-norm of a matrix* $\mathbf{A}$*, we always have*

$$\|\mathbf{A}\|_2 = \sqrt{\lambda_{\max}(\mathbf{A}^\top\mathbf{A})} \tag{19}$$

Proof. By the definition of 2-norm, we have

$$\|\mathbf{A}\|_2 = \sup_{\mathbf{x}\neq\mathbf{0}} \frac{\|\mathbf{A}\mathbf{x}\|_2}{\|\mathbf{x}\|_2}. \tag{20}$$

We first prove that there exits a vector $\mathbf{v}$ such that

$$\frac{\|\mathbf{A}\mathbf{v}\|_2}{\|\mathbf{v}\|_2} = \sqrt{\lambda_{\max}(\mathbf{A}^\top\mathbf{A})} \tag{21}$$

Consider the decomposition of $\mathbf{A}^\top\mathbf{A}$ which is a PSD, we always have

$$\mathbf{A}^\top\mathbf{A} = \mathbf{U}^\top \operatorname{diag}(\lambda_1, \cdots, \lambda_n)\mathbf{U}, \tag{22}$$

where $\mathbf{U}$ is orthonormal matrix which is full-rank. So we can find a $\mathbf{v}$ such that

$$\mathbf{U}\mathbf{v} = \begin{bmatrix} 1 \\ 0 \\ \vdots \\ 0 \end{bmatrix} \tag{23}$$

which means

$$\|\mathbf{A}\mathbf{v}\|_2 = \sqrt{\mathbf{v}^\top\mathbf{A}^\top\mathbf{A}\mathbf{v}} = \sqrt{(\mathbf{U}\mathbf{v})^\top \operatorname{diag}(\lambda_1, \cdots, \lambda_n)(\mathbf{U}\mathbf{v})} = \sqrt{\lambda_1} = \sqrt{\lambda_1}\|\mathbf{v}\|_2. \tag{24}$$

So, we find a $\mathbf{v}$ satisfies (21) which means:

$$\sqrt{\lambda_1} = \frac{\|\mathbf{A}\mathbf{v}\|_2}{\|\mathbf{v}\|_2} \leq \sup_{\mathbf{x}\neq\mathbf{0}} \frac{\|\mathbf{A}\mathbf{x}\|_2}{\|\mathbf{x}\|_2} = \|\mathbf{A}\|_2. \tag{25}$$

For all $\mathbf{u} \in \{\mathbf{x} : \mathbf{x} \neq \mathbf{0}\}$, we have

$$\mathbf{u}^\top\mathbf{A}^\top\mathbf{A}\mathbf{u} = (\mathbf{U}\mathbf{u})^\top \operatorname{diag}(\lambda_1, \cdots, \lambda_n)(\mathbf{U}\mathbf{u}) = \sum_i \lambda_i(\mathbf{U}\mathbf{u})_i^2 \tag{26}$$

$$\leq \lambda_1 \sum_i (\mathbf{U}\mathbf{u})_i^2 = \lambda_1\|\mathbf{U}\mathbf{u}\|^2 = \lambda_1\mathbf{u}^\top\mathbf{U}^\top\mathbf{U}\mathbf{u} = \lambda_1\|\mathbf{u}\|^2 \tag{27}$$

which means

$$\frac{\|\mathbf{A}\mathbf{u}\|^2}{\|\mathbf{u}\|^2} \leq \lambda_1 \tag{28}$$

Taking the sup in both side we have

$$\sup_{\mathbf{x}\neq\mathbf{0}} \frac{\|\mathbf{A}\mathbf{x}\|}{\|\mathbf{x}\|} \leq \sqrt{\lambda_1}. \tag{29}$$

Combining (25) and (29), we finish the proof. □

3 Basic Multi-Variable Function Analysis

3.1 Derivative

Consider a function $f(\mathbf{x}) : \mathbb{R}^d \to \mathbb{R}^1$, the gradient of f is

$$\nabla f(\mathbf{x}) = \begin{bmatrix} \frac{\partial f(\mathbf{x})}{\partial x_1} \\ \vdots \\ \frac{\partial f(\mathbf{x})}{\partial x_d} \end{bmatrix} \in \mathbb{R}^d. \tag{30}$$

Consider a function $\mathbf{F}(\mathbf{x}) = \begin{bmatrix} F_1(\mathbf{x}) \\ \vdots \\ F_m(\mathbf{x}) \end{bmatrix} : \mathbb{R}^d \to \mathbb{R}^m$, The derivative/Jacobi Matrix of $\mathbf{F}$ is

$$d_{\mathbf{F}}(\mathbf{x}) = \mathbf{J}_{\mathbf{F}}(\mathbf{x}) = \frac{\partial \mathbf{F}(\mathbf{x})}{\partial \mathbf{x}} = \begin{bmatrix} \nabla F_1(\mathbf{x})^\top \\ \nabla F_2(\mathbf{x})^\top \\ \vdots \\ \nabla F_m(\mathbf{x})^\top \end{bmatrix} \in \mathbb{R}^{m \times d} \tag{31}$$

If we choose $\mathbf{F}(\mathbf{x}) = \nabla f(\mathbf{x}) : \mathbb{R}^d \to \mathbb{R}^d$ ($F_i(\mathbf{x}) = \nabla_{x_i} f(\mathbf{x})$), the Hessian matrix of f is defined to be

$$\nabla^2 f(\mathbf{x}) = \mathbf{J}_{\nabla f}(\mathbf{x}) = \left[\frac{\partial^2 f(\mathbf{x})}{\partial x_i \partial x_j} \right]_{ij} \in \mathbb{R}^{d \times d} \tag{32}$$

Rules for calculating the derivative (these rules can be treated as a generation for the rule in Single Variable Calculus

- $d_{\mathbf{h}^\top \mathbf{g}}(\mathbf{x}) = \mathbf{h}(\mathbf{x})^\top d_{\mathbf{g}}(\mathbf{x}) + \mathbf{g}(\mathbf{x})^\top d_{\mathbf{h}}(\mathbf{x})$
- (Chain's Rule) $d_{\mathbf{h} \circ \mathbf{g}}(\mathbf{x}) = d_{\mathbf{h}}(\mathbf{g}(\mathbf{x})) d_{\mathbf{g}}(\mathbf{x})$

Questions

1. $\mathbf{F}(\mathbf{x}) = \mathbf{A}\mathbf{x}$: $\frac{\partial \mathbf{F}(\mathbf{x})}{\partial \mathbf{x}} = \mathbf{A}$
2. $f(\mathbf{x}) = \mathbf{a}^\top \mathbf{x}$: $\nabla f(\mathbf{x}) = \mathbf{a}$ and $\frac{\partial f(\mathbf{x})}{\partial \mathbf{x}} = [\nabla f(\mathbf{x})^\top] = \mathbf{a}^\top$
3. $f(\mathbf{x}) = \mathbf{x}^\top \mathbf{A}\mathbf{x}$:

$$d_f(\mathbf{x}) = \frac{\partial (\mathbf{A}^\top \mathbf{x})^\top \mathbf{x}}{\partial \mathbf{x}} = (\mathbf{A}^\top \mathbf{x})^\top \frac{\partial \mathbf{x}}{\partial \mathbf{x}} + \mathbf{x}^\top \frac{\partial (\mathbf{A}^\top \mathbf{x})}{\partial \mathbf{x}} = \mathbf{x}^\top \mathbf{A} + \mathbf{x}^\top \mathbf{A}^\top \tag{33}$$

$$\nabla f(\mathbf{x}) = [d_f(\mathbf{x})]^\top = (\mathbf{A} + \mathbf{A}^\top)\mathbf{x} \tag{34}$$

$$\nabla^2 f(\mathbf{x}) = d_{\nabla f}(\mathbf{x}) = \mathbf{A} + \mathbf{A}^\top \tag{35}$$

If $\mathbf{A}$ is symmetric, then $\nabla f(\mathbf{x}) = 2\mathbf{A}\mathbf{x}$

4. $f(\mathbf{x}) = \|\mathbf{A}\mathbf{x} - \mathbf{b}\|^2$ (we let $\mathbf{u} = \mathbf{A}\mathbf{x} - \mathbf{b}$)

$$d_f(\mathbf{x}) = \frac{\partial \|\mathbf{u}\|^2}{\partial \mathbf{u}} \frac{\partial \mathbf{u}}{\partial \mathbf{x}} = 2\mathbf{u}^\top \mathbf{A} = 2(\mathbf{A}\mathbf{x} - \mathbf{b})^\top \mathbf{A} \tag{36}$$

$$\nabla f(\mathbf{x}) = [d_f(\mathbf{x})]^\top = 2\mathbf{A}^\top (\mathbf{A}\mathbf{x} - \mathbf{b}) \tag{37}$$

$$\nabla^2 f(\mathbf{x}) = d_{\nabla f}(\mathbf{x}) = 2\mathbf{A}^\top \mathbf{A} \tag{38}$$

5. $f(\mathbf{x}) = \log\left(1 + \mathbf{e}^{-\mathbf{a}^\top \mathbf{x}}\right)$:

$$d_f(\mathbf{x}) = -\frac{1}{1+\mathbf{e}^{-\mathbf{a}^\top \mathbf{x}}}\mathbf{e}^{-\mathbf{a}^\top \mathbf{x}}\mathbf{a}' \tag{39}$$

$$\nabla f(\mathbf{x}) = [d_f(\mathbf{x})]^\top = -\left(\frac{\mathbf{e}^{-\mathbf{a}^\top \mathbf{x}}}{1+\mathbf{e}^{-\mathbf{a}^\top \mathbf{x}}}\right)\mathbf{a} \tag{40}$$

$$\nabla^2 f(\mathbf{x}) = p(1-p)\mathbf{a}\mathbf{a}' \tag{41}$$

where $p = \left(\frac{1}{1+\mathbf{e}^{-\mathbf{a}^\top \mathbf{x}}}\right)$.

3.2 Integral*

The following proposition will be useful to obtain the properties of a convex function, you can skip them first.

Proposition 3.1.

$$f(\mathbf{y}) - f(\mathbf{x}) = \int_{t=0}^{1} \nabla f(\mathbf{x} + t(\mathbf{y}-\mathbf{x}))^\top (\mathbf{y}-\mathbf{x})dt \tag{42}$$

$$\nabla f(\mathbf{y}) - \nabla f(\mathbf{x}) = \int_{t=0}^{1} \nabla^2 f(\mathbf{x} + t(\mathbf{y}-\mathbf{x}))(\mathbf{y}-\mathbf{x})dt \tag{43}$$

3.3 Taylor Series Expansion

We only consider the first-order Taylor series expansion and second-order Taylor series expansion. Higher-order Taylor expansion is used in higher-order(tensor-order) optimization method which are not covered in this course.

Definition 3.2 (**The first-order Taylor expansion** at $\mathbf{x}$)**.**

$$f_1(\mathbf{y};\mathbf{x}) \overset{def}{=} f(\mathbf{x}) + \langle \nabla f(\mathbf{x}), \mathbf{y}-\mathbf{x}\rangle, \tag{44}$$

Definition 3.3 (**The Second-order Taylor expansion** at $\mathbf{x}$)**.**

$$f_2(\mathbf{y};\mathbf{x}) \overset{def}{=} f(\mathbf{x}) + \langle \nabla f(\mathbf{x}), \mathbf{y}-\mathbf{x}\rangle + \frac{1}{2}(\mathbf{y}-\mathbf{x})^\top \nabla^2 f(\mathbf{x})(\mathbf{y}-\mathbf{x}). \tag{45}$$

We now consider the error between $f(\mathbf{y})$ and $f_1(\mathbf{y};\mathbf{x})$, $f_2(\mathbf{y};\mathbf{x})$.

Proposition 3.4. *If f is differentiable, then*

$$\epsilon_1(\mathbf{y};\mathbf{x}) \overset{def}{=} f_1(\mathbf{y};\mathbf{x}) - f(\mathbf{x}) \tag{46}$$

satisfies that

$$\lim_{\mathbf{y}\to\mathbf{x}} \frac{\epsilon_1(\mathbf{y};\mathbf{x})}{\|\mathbf{y}-\mathbf{x}\|} = 0 \tag{47}$$

Proposition 3.5. *If f is twice-differentiable, then*

$$\epsilon_2(\mathbf{y};\mathbf{x}) \overset{def}{=} f_2(\mathbf{y};\mathbf{x}) - f(\mathbf{x}) \tag{48}$$

satisfies that

$$\lim_{\mathbf{y}\to\mathbf{x}} \frac{\epsilon_2(\mathbf{y};\mathbf{x})}{\|\mathbf{y}-\mathbf{x}\|^2} = 0 \tag{49}$$

4 Basic Knowledge for Convex Function

Most results in this section can be found in Nesterov's lecture note on convex optimization Nesterov et al. [2018 `https://link.springer.com/book/10.1007/978-3-319-91578-4`].

Definition 4.1 (**Convex Set**). *A set $\mathcal{S} \subseteq \mathbb{R}^d$ is called a convex set if for all $\mathbf{x}$, $\mathbf{y} \in \mathcal{S}$ and $\theta \in [0,1]$*

$$\theta\mathbf{x} + (1-\theta)\mathbf{y} \in \mathcal{S}. \tag{50}$$

Definition 4.2 (**Convex function**). *f is a convex function if*

$$f(\theta\mathbf{x} + (1-\theta)\mathbf{y}) \leq \theta f(\mathbf{x}) + (1-\theta) f(\mathbf{y}) \tag{51}$$

for all $\theta \in [0,1]$ and $\mathbf{x}, \mathbf{y} \in \mathrm{dom}(f)$ and $\mathrm{dom}(f)$ is a convex set.

Proposition 4.3. *The sublevel set of a convex function f*

$$\mathcal{S} = \{\mathbf{x} | f(\mathbf{x}) \leq 0\} \tag{52}$$

is a convex set

Proof.

$$f(\theta\mathbf{x} + (1-\theta)\mathbf{y}) \leq \theta f(\mathbf{x}) + (1-\theta) f(\mathbf{y}) \leq 0 \tag{53}$$

□

Proposition 4.4. *If f is twice differentiable, the following statements*

- *(a). f is convex*
- *(b). $f(\mathbf{y}) \geq f(\mathbf{x}) + \langle \nabla f(\mathbf{x}), \mathbf{y} - \mathbf{x} \rangle$*
- *(c). $\nabla^2 f(\mathbf{x}) \succeq \mathbf{0}$*

are equal

Proof. We first prove $(a) \Leftrightarrow (b)$.
(a) $\Rightarrow$ (b): for all $\theta \in [0,1]$ and $\mathbf{x}_\theta = \theta\mathbf{x} + (1-\theta)\mathbf{y}$, we have for all θ

$$f(\mathbf{y}) \geq \frac{1}{1-\theta}(f(\mathbf{x}_\theta) - \theta f(\mathbf{x})) = f(\mathbf{x}) + \frac{f(\mathbf{x}_\theta) - f(\mathbf{x})}{1-\theta} \tag{54}$$

we then have

$$f(\mathbf{y}) - f(\mathbf{x}) \geq \lim_{\theta \to 1} \frac{f(\theta\mathbf{x} + (1-\theta)\mathbf{y}) - f(\mathbf{x})}{1-\theta} = \lim_{\theta \to 1} \frac{f(\mathbf{x} + (1-\theta)(\mathbf{y} - \mathbf{x})) - f(\mathbf{x})}{1-\theta} \tag{55}$$

$$= \lim_{\alpha \to 0} \frac{f(\mathbf{x} + \alpha(\mathbf{y} - \mathbf{x})) - f(\mathbf{x})}{\alpha} = \langle \nabla f(\mathbf{x}), \mathbf{y} - \mathbf{x} \rangle \tag{56}$$

(b) $\Rightarrow$ (a): we have

$$f(\mathbf{x}_\theta) \leq f(\mathbf{y}) - \theta \langle \nabla f(\mathbf{x}_\theta), \mathbf{y} - \mathbf{x} \rangle \tag{57}$$

$$f(\mathbf{x}_\theta) \leq f(\mathbf{x}) + (1-\theta) \langle \nabla f(\mathbf{x}_\theta), \mathbf{y} - \mathbf{x} \rangle \tag{58}$$

We then prove $(b) \Leftrightarrow (c)$.
(b) $\Rightarrow$ (c) for all $\mathbf{y}$, we have

$$f(\mathbf{y}) - f(\mathbf{x}) \geq \langle \nabla f(\mathbf{x}), \mathbf{y} - \mathbf{x} \rangle \tag{59}$$

Please refer to Theorem 2.1.4 in Nesterov's book Nesterov et al. [2018 `https://link.springer.com/book/10.1007/978-3-319-91578-4`]
(c) $\Rightarrow$ (b)

$$\begin{aligned} f(\mathbf{y}) - f(\mathbf{x}) - \langle \nabla f(\mathbf{x}), \mathbf{y} - \mathbf{x} \rangle &= \int_{t=0}^{1} \langle \nabla f(\mathbf{x} + t(\mathbf{y} - \mathbf{x})) - \nabla f(\mathbf{x}), \mathbf{y} - \mathbf{x} \rangle dt && (60) \\ &= \int_{t=0}^{1} \langle \int_{s=0}^{1} \nabla^2 f(\mathbf{x} + ts(\mathbf{y} - \mathbf{x}))t(\mathbf{y} - \mathbf{x}) ds, \mathbf{y} - \mathbf{x} \rangle dt && (61) \\ &= \int_{t=0}^{1} t(\mathbf{y} - \mathbf{x})^\top \left(\int_{s=0}^{1} \nabla^2 f(\mathbf{x} + ts(\mathbf{y} - \mathbf{x})) ds \right) (\mathbf{y} - \mathbf{x}) dt \geq 0 && (62) \end{aligned}$$

□

Definition 4.5 (μ**-strongly convex function**). *f is a μ-strongly convex function if $f(\mathbf{x}) - \frac{\mu}{2}\|\mathbf{x}\|^2$ is a convex function.*

Proposition 4.6. *If f is twice differentiable, the following statements*

- *(a). f is μ-strongly convex*
- *(b). $f(\mathbf{y}) \geq f(\mathbf{x}) + \langle \nabla f(\mathbf{x}), \mathbf{y} - \mathbf{x} \rangle + \frac{\mu}{2}\|\mathbf{y} - \mathbf{x}\|^2$*
- *(c). $\nabla^2 f(\mathbf{x}) \succeq \mu \mathbf{I}$*

all equal.

Proposition 4.7. * *If f is μ-strongly convex, then f enjoys the following properties:*

$$f(\mathbf{x}) \leq f(\mathbf{y}) + \langle \nabla f(\mathbf{y}), \mathbf{x} - \mathbf{y} \rangle + \frac{1}{2\mu}\|\nabla f(\mathbf{x}) - \nabla f(\mathbf{y})\|^2 \tag{63}$$

$$\mu\|\mathbf{x} - \mathbf{y}\|^2 \leq \langle \nabla f(\mathbf{x}) - \nabla f(\mathbf{y}), \mathbf{x} - \mathbf{y} \rangle \tag{64}$$

$$\mu\|\mathbf{x} - \mathbf{y}\| \leq \|\nabla f(\mathbf{x}) - \nabla f(\mathbf{y})\| \tag{65}$$

Definition 4.8 (l**-smooth function**). *f is l-smooth if*

$$\|\nabla f(\mathbf{x}) - \nabla f(\mathbf{y})\| \leq l\|\mathbf{x} - \mathbf{y}\| \tag{66}$$

Questions

1. If $f_i(\mathbf{x})$ are all convex function, then $g(\mathbf{x}) \stackrel{\text{def}}{=} \max_i\{f_i(\mathbf{x})\}$ is convex function.

   Proof.

$$\begin{aligned} g(\theta\mathbf{x} + (1-\theta)\mathbf{y}) &= \max_i\{f_i(\theta\mathbf{x} + (1-\theta)\mathbf{y})\} && (67) \\ &\leq \max_i\{\theta f_i(\mathbf{x}) + (1-\theta) f_i(\mathbf{y})\} && (68) \\ &\leq \theta \max_i\{f_(\mathbf{x})\} + (1-\theta)\max_i\{f_i(\mathbf{y})\} && (69) \\ &= \theta g(\mathbf{x}) + (1-\theta) g(\mathbf{y}) && (70) \end{aligned}$$

   □

2. If $f_i(\mathbf{x})$ are all concave functions, then $g(\mathbf{x}) \stackrel{\text{def}}{=} \min_i\{f_i(\mathbf{x})\}$ is concave function.

5 Convex Optimization

$$\begin{aligned} \min_{\mathbf{x}} \ & f(\mathbf{x}) \\ \text{Subject to} \ & \mathbf{x} \in \mathcal{S}, \end{aligned} \tag{71}$$

we call f the object function and $\mathcal{S} \subseteq \mathbb{R}^d$ is the feasible set.

Definition 5.1 (**Convex Optimization Problem**). *Both the object function and the feasible set are convex.*

Definition 5.2 (**Standard form of convex optimization**).

$$\min_{\mathbf{x}\in\mathbb{R}^d} \quad f(\mathbf{x}) \tag{72}$$

$$\text{subject to} \quad g_i(\mathbf{x}) \leq 0 \ \ \textit{for all} \ \ i \tag{73}$$

$$\mathbf{A}\mathbf{x} = \mathbf{b} \tag{74}$$

where f and $\mathbf{g}_i$ are convex function.

Definition 5.3 (**Global minimum**). *We call a $\mathbf{x}^* \in \mathcal{S}$ a global minimum if*

$$\mathbf{x}^* = \arg\min_{\mathbf{x}\in\mathcal{S}} f(\mathbf{x}) \tag{75}$$

Definition 5.4 (**Local minimum**). *: We call $\mathbf{x}^* \in \mathcal{S}$ a local minimum if*

$$\mathbf{x}^* = \arg\min_{\mathbf{x}\in\mathcal{S}\cap\{\mathbf{x}|\|\mathbf{x}-\mathbf{x}^*\|\leq\epsilon\}} f(\mathbf{x}) \tag{76}$$

Proposition 5.5. *For convex optimization problem, every local optimum is also a global optimum.*

5.1 Linear Programming

Definition 5.6 (**Linear Programming**). *The object function and constraint functions are all affine functions. The LP problem can be written as*

$$\min_{\mathbf{x}\in\mathbb{R}^d} \quad \mathbf{c}^\top\mathbf{x} \tag{77}$$

$$\text{Subject to:} \quad \mathbf{A}\mathbf{x} \leq \mathbf{b} \tag{78}$$

$$\mathbf{x} \geq \mathbf{0} \tag{79}$$

Examples: Transportation optimization problem, manufacturing optimization problem, Sorting tasks.

Manufacturing task:

- c_i profit per unit of product i
- b_j available units of Raw material j
- $a_{i,j}$ units of Raw material j for one unit of Product i

$$\min_{\mathbf{x}\in\mathbb{R}^n} \quad \mathbf{x}^\top\mathbf{c} \tag{80}$$

$$\text{Subject to:} \quad \mathbf{a}_{:,j}^\top\mathbf{x} \leq b_j \ \ \text{for all } j = 1, \cdots, m \tag{81}$$

$$\mathbf{x} \geq \mathbf{0} \tag{82}$$

References

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